

G2 Differentiating functions

Name: _____

Class: _____

Date: _____

Time: **113 minutes**

Marks: **92 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A particle moves in a straight line and at time t seconds has velocity $v \text{ m s}^{-1}$, where

$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) The mass of the particle is 3 kg.

Find the resultant force on the particle when $t = 4$.

(2)

(Total 4 marks)

Q2.

A particle moves in a straight line and at time t seconds has velocity $v \text{ m s}^{-1}$, where

$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) The mass of the particle is 3 kg.

Find the resultant force on the particle when $t = 4$.

(2)

- (c) When $t = 0$, the displacement of the particle from the origin is 5 metres.

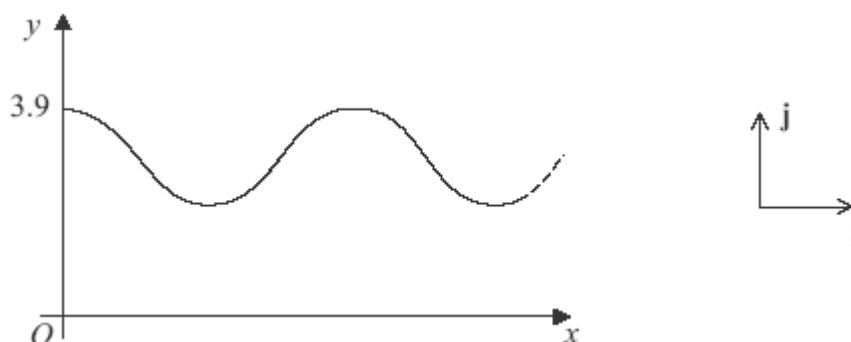
Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 8 marks)

Q3.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = 1.2t \mathbf{i} + (3 + 0.9 \cos t) \mathbf{j}$$

where the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.

- (a) Find an expression for Jane's velocity at time t . (2)
- (b) (i) Find an expression for Jane's speed at time t . (2)
- (ii) Find Jane's maximum speed during the ride. (2)
- (Total 6 marks)**

Q4.

A particle moves in a straight line and at time t has velocity v , where

$$v = 2t - 12e^{-t}, \quad t \geq 0$$

- (a) (i) Find an expression for the acceleration of the particle at time t . (2)
- (ii) State the range of values of the acceleration of the particle. (3)
- (b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t . (4)

(Total 9 marks)

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Q5.

A particle P , of mass 0.25 kg, moves in a horizontal plane so that at time t seconds its velocity $\mathbf{v}$ m s⁻¹ is given by

$$\mathbf{v} = 8 \sin t \mathbf{i} + 4 \cos t \mathbf{j}$$

- (a) Find the force, $\mathbf{F}$ newtons, acting on P at time t . (3)
- (b) (i) Show that the magnitude of $\mathbf{F}$ at time t is

$$\sqrt{(3 \cos^2 t + 1)}$$

(3)

- (ii) Hence state the range of values of the magnitude of $\mathbf{F}$.

(2)

(Total 8 marks)

Q6.

A particle P moves in a horizontal plane so that at time t seconds after the motion begins its position vector $\mathbf{r}$ metres is given by

$$\mathbf{r} = \sin 2t \mathbf{i} + 6t \mathbf{j}$$

- (a) (i) Find the velocity of P at time t .

(2)

- (ii) Show that the speed of P at time t is

$$\sqrt{(4 \cos^2 2t + 36)}$$

(3)

- (iii) Find the first value of t for which the speed of P has its minimum value.

(2)

- (b) (i) The mass of P is 0.25 kg. Find the force, $\mathbf{F}$ newtons, acting on P at time t .

(3)

- (ii) State the direction of $\mathbf{F}$ and explain why its magnitude never exceeds one newton.

(2)

(Total 12 marks)

Q7.

A particle moves in a straight line so that at time t seconds it has velocity v m s⁻¹, where

$$v = 3t^2 - 5t.$$

- (a) Find an expression for the acceleration of the particle at time t .

(2)

- (b) Find the value of t when the acceleration is zero.

(1)

(Total 3 marks)

Q8.

A particle moves so that, at time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 4e^{-t}\mathbf{i} + (6t + 3e^{-t})\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Show that the velocity of the particle is $(-4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when $t = 0$. (3)
- (b) Find an expression for the acceleration of the particle at time t . (2)
- (c) Find the magnitude of the acceleration of the particle when $t = 0$. (2)
- (d) Describe what happens to the velocity of the particle as t becomes large. (2)
- (Total 9 marks)**

Q9.

A cyclist moves from rest along a straight horizontal road. At time t seconds, the displacement of the cyclist from his initial position is s metres.

- (a) For $0 \leq t \leq 10$,

$$s = \frac{t^4}{400} - \frac{t^3}{10} + \frac{3t^2}{2}$$

- (i) Find s when $t = 10$. (1)
- (ii) Find the velocity of the cyclist when $t = 10$. (3)
- (iii) Find the acceleration of the cyclist when $t = 10$. (3)
- (b) For $t \geq 10$ the cyclist moves with a constant velocity, so that

$$s = ht - k$$

where h and k are constants. Find the values of h and k .

(3)
(Total 10 marks)

Q10.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = (4t - 0.01t^2)\mathbf{i} + (5 - 3t - 0.04t^2)\mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) State the position of the boat when $t = 0$. (1)

- (b) Find the time when the boat is due south of its initial position found in part (a).

(4)

- (c) Find the time when the boat is travelling south east.

(5)

(Total 10 marks)

Q11.

A particle moves along a straight line. At time t the displacement of the particle from its initial position is x , where

$$x = 4t + 2e^{-t} - 2$$

- (a) Find the velocity of the particle at time t .

(2)

- (b) Find the acceleration of the particle at time t .

(2)

- (c) Describe what happens to the acceleration of the particle as t increases.

(2)

(Total 6 marks)

Q12.

A boat moves so that its position vector, $\mathbf{r}$ metres, at time t seconds is

$$\mathbf{r} = 40 \cos\left(\frac{t}{20}\right) \mathbf{i} + 80 \sin\left(\frac{t}{20}\right) \mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the boat at time t .

(3)

- (b) In what direction is the boat travelling when $t = 0$? Justify your answer.

(2)

- (c) At what time is the boat travelling due south for the first time?

(2)

(Total 7 marks)